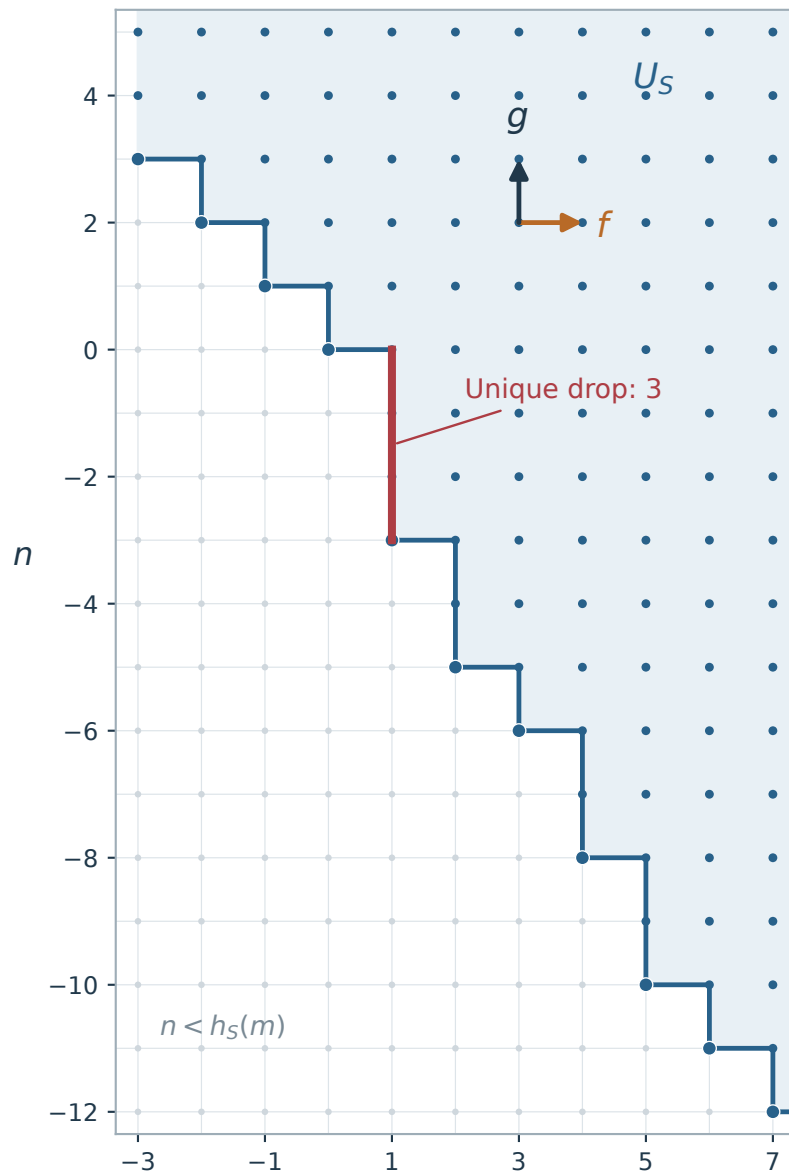
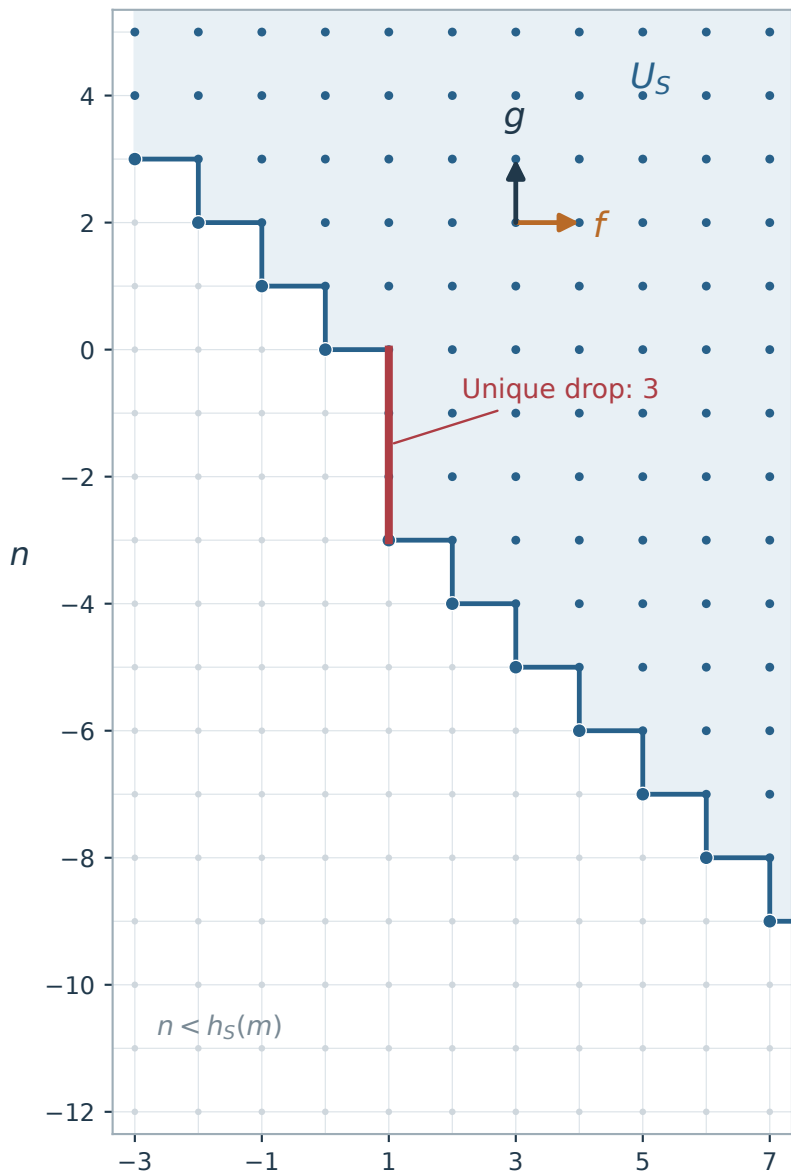


Two commuting injections on coded staircases

$$U_S = \{(m, n) \in \mathbb{Z}^2 : n \geq h_S(m)\} \quad \cdot \quad f(m, n) = (m+1, n) \quad \cdot \quad g(m, n) = (m, n+1)$$

$$S = \emptyset$$

$$S = \{0, 2, 3\}$$



m Points of U_S
 Region above the boundary
 Marker $d_S(0) = 3$